



Scratch — Pen Tool

Let's Draw with Code!

Grade 2 Lesson · Intro to the Pen Extension

scratch.mit.edu | Colourful blocks — no typing needed!

Code * Create * Share

What is Scratch?

Scratch is a fun coding platform where you can create stories, games, and amazing art! It uses **colourful drag-and-drop blocks** — no typing needed!

Game s	Make animations and games Build interactive projects with characters (sprites) and backgrounds.
Draw	Draw pictures using code Use the Pen extension to create art automatically.
Share	Share your creations Publish to the Scratch community and remix others' work.

Meet the Pen Tool

The **Pen tool** lets you draw lines and shapes by coding! When your sprite moves, it leaves a trail — like a magic pencil on screen.

Dr aw	Draw lines as your sprite moves across the stage
Co lor	Change the pen color and size whenever you like
Lif t	Lift the pen (Pen Up) to move without drawing a line

How Does the Pen Tool Work?

Think of your sprite as holding a real pencil. When the pen is **down**, it draws. When the pen is **up**, it moves freely without leaving a mark.

Pe n Do wn	Pen Down — the sprite starts drawing as it moves
Pe n Up	Pen Up — the sprite stops drawing (lifts the pen!)
Co lor	Set Pen Color — pick any colour you like

Try
it!

Move your sprite and watch the magic happen!

Pen Tool Blocks in Scratch

After adding the Pen extension, you'll find these blocks in your palette:

pen down	Start drawing — the sprite leaves a trail as it moves
pen up	Stop drawing — move without leaving a mark
set pen color	Choose any colour for your pen
change pen size	Make lines thick or thin
clear	Erase all drawings on the stage
stamp	Copy the sprite's image onto the stage
set pen shade	Adjust the brightness of the pen colour
change pen hue	Shift the colour tone of the pen

Getting Started — 3 Easy Steps

- 1 Open Scratch and add the Pen Extension**
Go to *scratch.mit.edu* → open a new project → click the blue **Extensions** button (bottom left) → choose **Pen**. Pen blocks now appear in your palette!
- 2 Drag the 'Pen Down' block into your script**
Find **pen down** in the Pen section → drag it under your **when flag clicked** start block. Your sprite is now ready to draw!
- 3 Move your sprite and watch it draw!**
Add a **move 100 steps** block. Click the green flag — your sprite glides and leaves a colourful trail. Try moving in different directions to see your drawing grow!

Create Your Own Shape

Use **move** and **turn** blocks together with the pen to draw these shapes:

 Square Move & turn 4 times (90 deg)	 Triangle Move & turn 3 times (120 deg)
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**Circle**

Repeat: small move + tiny turn

**Star**

Combine moves + pen commands

■ **Pro tip:** Use a **repeat** loop block to draw shapes without repeating the same blocks over and over!

Mini Project: Draw a House ■

Follow these steps to code your very own house in Scratch!

1	Pen down — draw a square for the house base
2	Draw a triangle on top for the roof
3	Add lines for windows and a door
4	Change pen colour for fun details
5	Add grass or a sky background
6	Experiment with different pen sizes
7	Add a sun or fluffy clouds
8	Try using your favourite colours
9	Save your amazing creation
10	Share your house with friends!

Tips & Tricks

Tip 1	Use Pen Up when moving to a new position without drawing
Tip 2	Change colours and sizes for creative effects
Tip 3	Use repeat loop blocks to make drawing easier and faster
Tip 4	Have fun experimenting — there's no wrong answer!
Tip 5	Save your project often so you don't lose your work

Quick Quiz! ■

Q1: What block makes your sprite START drawing?

- A) Pen Up
- B) Pen Down ✓**
- C) Move

Q2: Which block STOPS the sprite from drawing?

- A) Pen Down
- B) Set Pen Color
- C) Pen Up ✓**

Q3: Which block changes the line colour?

- A) Set Pen Color ✓**
- B) Pen Down
- C) Move

Correct answers are shown in **green bold**.

Summary: What We Learned Today

O K	Scratch is a fun, block-based coding tool anyone can use
O K	The Pen tool draws lines as your sprite moves
O K	Pen Down starts drawing; Pen Up stops drawing
O K	You can create squares, triangles, circles, and stars with code
O K	Change pen colour and size for creative effects
O K	Use repeat (loop) blocks to draw shapes more easily
O K	Practice by trying the mini-project: Draw a House!
O K	Keep experimenting, exploring, and sharing your creations

Keep Coding and Creating!

Every line of code you write is a step toward something incredible. You are a creative coder — keep going and have fun! Visit **scratch.mit.edu** to start your next project.